Probability

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MATH 240

Week 7, Lecture #2 — Ch. 3 of OpenIntro

Reminders

- Homework #5 due on Friday at 11:59pm
  - One file for textbook problems
  - One file for R code

Language of Probability

We need three components:

1. A random experiment
   - Some activity, process, or (actual or thought) experiment whose outcome is uncertain
   - e.g. rolling a die
2. A sample space
   - A set/collection of all possible outcomes from the random experiment, denoted as $\mathcal{S}$
   - e.g. $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ for rolling a die
3. An event
   - A collection of outcomes
   - e.g. Getting an odd number on a roll of a die

Activity: Components in Random Experiments

For each of these scenarios, determine the **sample space** and an **event** from the sample space.

1. Dealing a card from a set of playing cards
2. Analyzing the number of classes a student is currently taking
3. Playing a single turn in Monopoly (or any other board game)

Foundations of Probability

We use a **probability distribution** to list all possible outcomes of the random experiment and their associated probabilities.

Example:

Dice sum	2	3	4	5	6	7	8	9	10	11	12
Prob.	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Table: Probability Distribution of Sum of Two Rolled Dice

What Makes a Probability Distribution?

Three things to check:

1. Are the outcomes **disjoint**?
   - Disjoint: two outcomes/events cannot occur at the same time
2. Is each probability **between 0 and 1**?
3. Do all probabilities **sum up to 1**?

If **yes**, we have a valid probability distribution!

Activity: Probability Distributions?

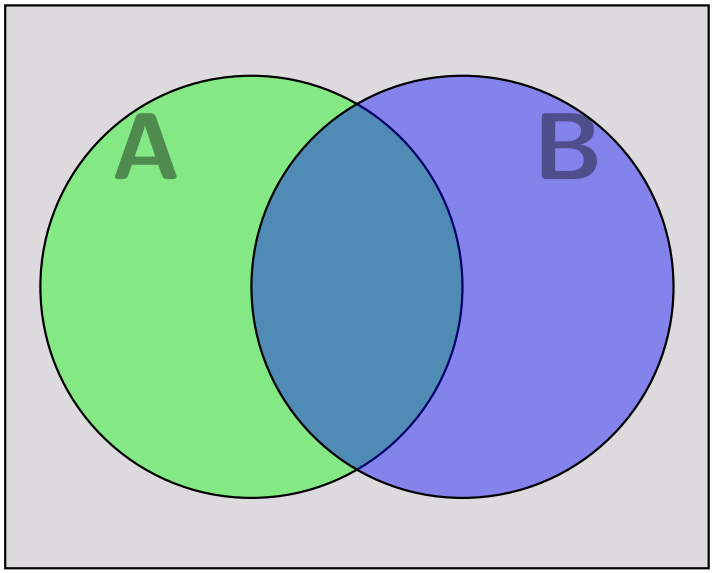
Determine if the following distributions are valid probability distributions:

1. I draw a card from a set of playing cards. I claim that there is a $\frac{1}{13}$ chance of getting some numeric face card, while there is a $\frac{2}{13}$ chance of getting a standard face card (e.g. A, J, Q, K).
2. There are 10 balls in an urn. 3 of them are red, 4 are blue, and 3 are green. I have $\frac{3}{10}$, $\frac{4}{10}$, and $\frac{3}{10}$ chances of getting a red, blue, or green ball respectively on the first draw.
3. I look at all possible household incomes and say that any American adults has this chance of making this annual income:
   - 20% between \$0-50k
   - 20% between \$50-80k
   - 20% between \$80-110k
   - 20% between \$110-200k
   - 20% between \$200-500k

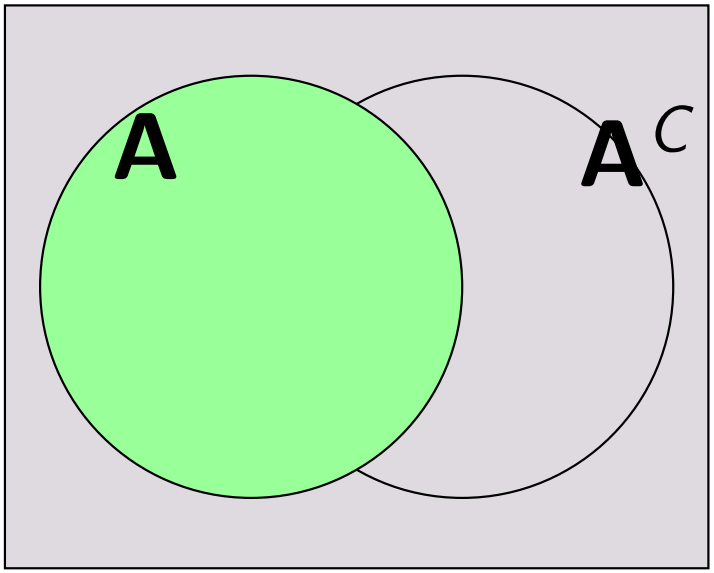
Probability - Events

Recall that an event is a collection of outcomes. Denote A and B as two separate events from the sample space. We can find the following:

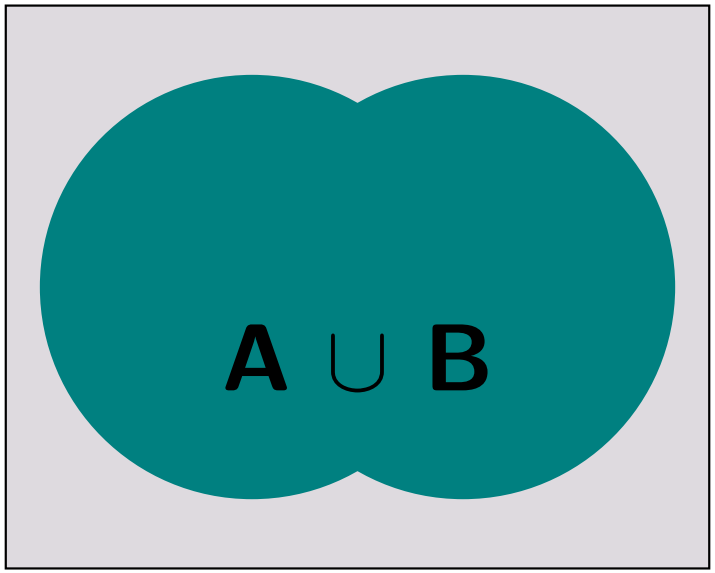
- The complement: A^C
  - All other outcomes not associated with event A in sample space
  - $P(A^C) = 1 - P(A)$
- The intersection: $A \cap B$
  - All outcomes associated with both events A and B
  - $P(A \cap B)$
- The union: $A \cup B$
  - All outcomes associated with events A, B, or both
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



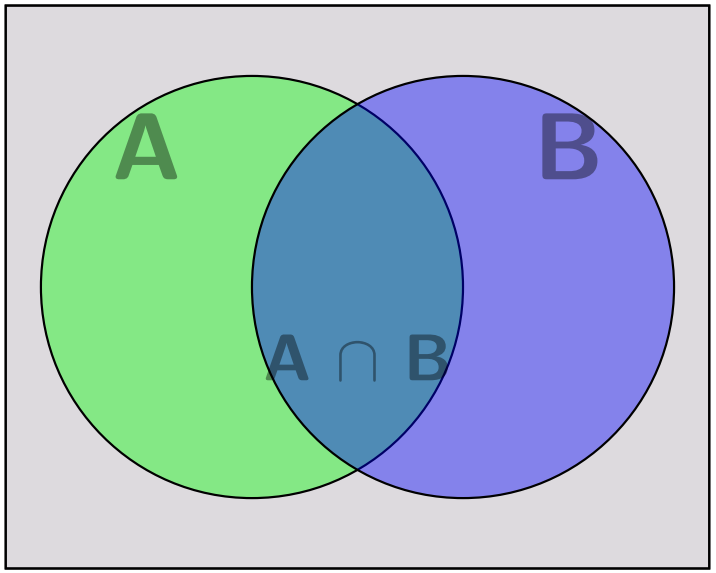
Events A and B in sample space.



Complement of A.



Union of A and B.



Intersection of A and B.

General Addition Rule

For events A and B that are disjoint, then:

$$P(A \cup B) = P(A) + P(B)$$

For events A and B that are not disjoint, then:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Activity: Two Dice

Suppose you roll two dice simultaneously. Let these events occur:

- A: Event of rolling a six
- B: Event of rolling a multiple of 3
- C: Event of rolling a 10 or greater

Find the following:

1. $P(A)$, $P(B)$, $P(C)$ individually
2. $P(A \cap B)$
3. $P(A \cup B)$